

Quality metrics in AI-Models' output: Simantic Similarity Score

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Where Semantic Quality Fits in AI Projects

- Content-generation tasks in products (chatbots, summarizers, recommender systems)
- Need for meaning-preserving outputs

How Semantic Similarity Works (Non-technical explanation)

- Measure underlying meaning
- Idea of text embeddings
- Meaning represented as vectors

BERT

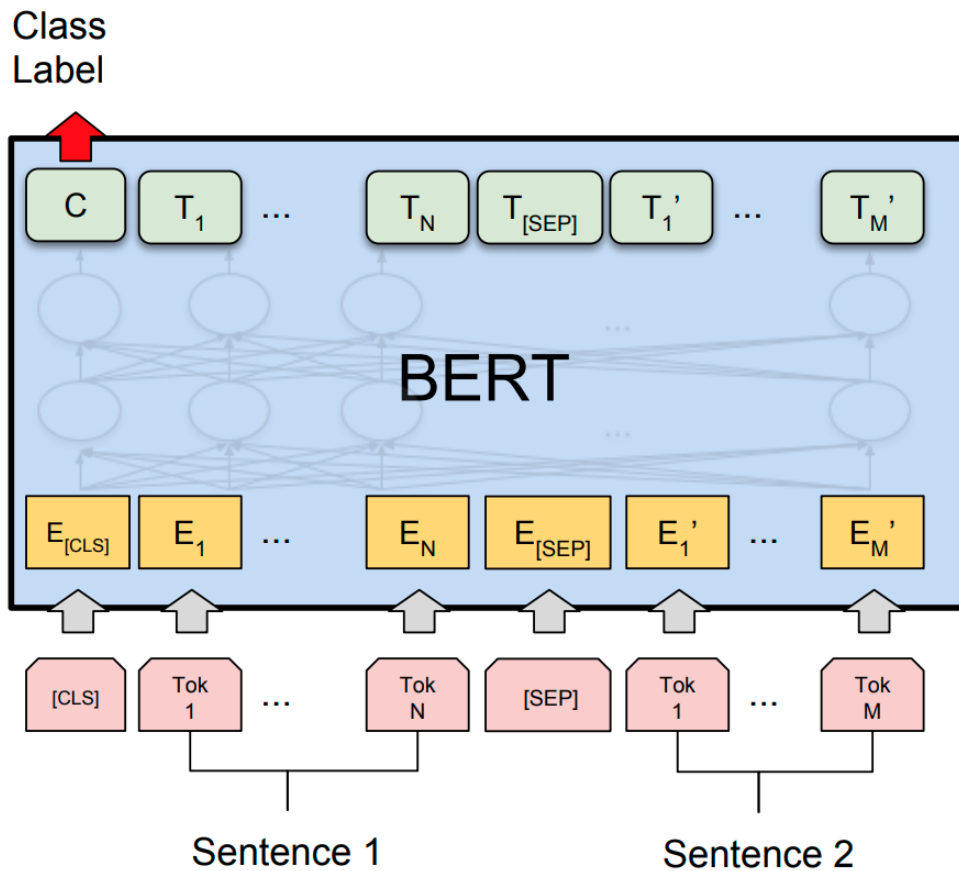
- Bidirectional Encoder Representations from Transformers
- published by researchers at Google AI Language.
- applying the bidirectional training of Transformer, a popular attention model to language modelling

Bidirectional and Attention model

- **Bidirectional:**
 - reads the text in both directions at the same time, left-to-right and right-to-left to understand full context.
 - This gives much better understanding
- **Attention model:**
 - is the mechanism that allows the system to look at all the words in a sentence at the same time and decide which words are most important for understanding each other.
 - It can weigh words differently depending on their importance, capturing long-range dependencies efficiently

Alternatives

- For Bidirectional Training:
 - Unidirectional models: GPT (Generative Pretrained Transformer) reads left-to-right only. It works well for generation tasks but may miss full sentence context for understanding.
- Attention Model
 - Shallow RNN / LSTM without attention: Can process sequences, but they treat all previous words equally or rely on hidden states, which may forget important information.
 - CNNs for text: Can detect local patterns but do not capture long-range dependencies effectively.



High-Level Example

- Sentence: “The movie was fantastic.”
- Tokenize: BERT splits the sentence:
[CLS] | the | movie | was | fantastic | [SEP]
- [CLS] represents the whole sentence,
- [SEP] marks the end of the sentence.
- BERT converts each token into something like this:

Each token becomes a vector

- BERT converts each token into something like this:
 - [CLS] $\rightarrow$ [0.12, 0.68, -0.24, ...]
 - the $\rightarrow$ [-0.03, 0.44, 0.10, ...]
 - movie $\rightarrow$ [0.56, 0.90, -0.12, ...]
 - was $\rightarrow$ [0.11, 0.03, 0.02, ...]
 - fantastic $\rightarrow$ [0.92, 0.33, 0.78, ...]
- Each list = a vector that represents meaning.
- Think of them as:
 - “movie” $\rightarrow$ a meaning-vector about films
 - “fantastic” $\rightarrow$ a meaning-vector about positivity

BERT mixes vectors and self-attention

- BERT looks at how the meaning-vectors relate:
- “fantastic” vector is strongly linked to “movie” vector
- “was” connects them
- “the” contributes little

Final [CLS] vector

- After processing:
 - [CLS] $\rightarrow$ [1.22, -0.44, 0.89, ...]
- This final [CLS] vector represents the meaning of the whole sentence.
- If the task is sentiment analysis, this vector tells BERT:
 - This sentence is positive.

Types and Alternatives

- RoBERTa & T5 → Higher accuracy but heavier computational cost.
- DistilBERT & ALBERT → Faster and lighter but may lose some nuance.
- ELECTRA → Efficient training but more complex setup.
- GPT models → Excellent for generation but expensive to fine-tune.

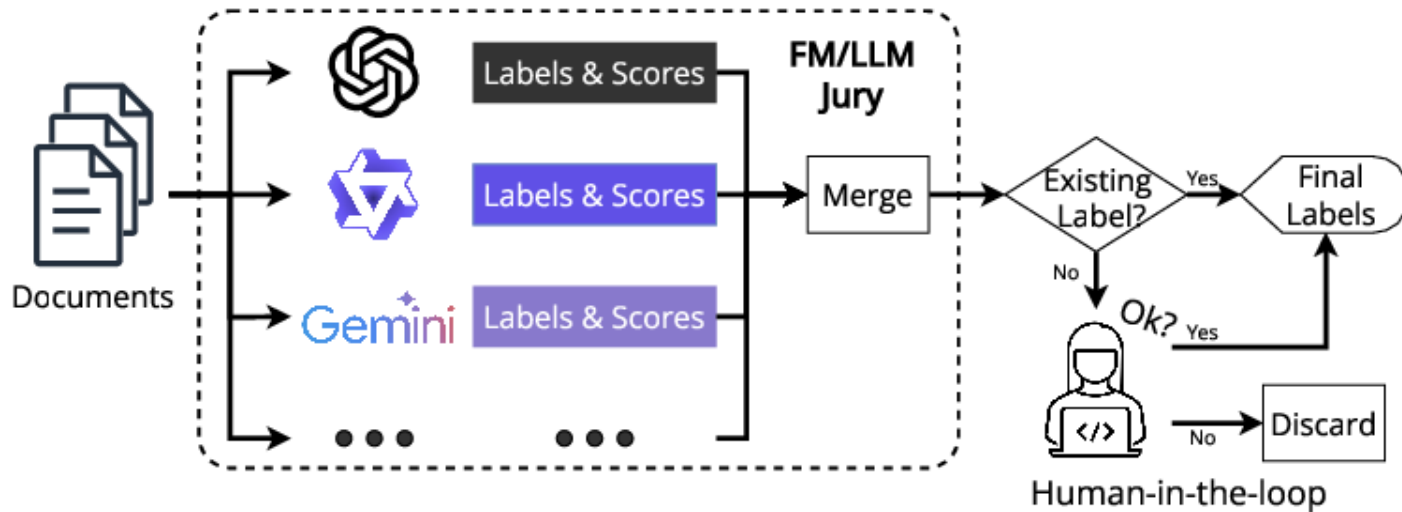
Using Type-4 Code Clone

- Type-4 clones are semantic or functional clones, meaning the code performs the same functionality but may have different implementations, variable names, or structure.
- Unlike Type-1 (exact) or Type-2 (renamed variables) or Type-3 (small edits), Type-4 focuses on behavioral equivalence rather than textual similarity.
- We can use Type-4 clone detection to identify whether different outputs are functionally equivalent, even if the code looks different.

LLMs as a Judge

- LLM-as-a-Judge uses LLMs to evaluate AI-generated texts based on custom criteria defined in an evaluation prompt.
- Treat an AI-Model as an automated evaluator that judges whether outputs (code or text) meet certain quality criteria.
- A prompt asking it to judge: e.g., “Does this candidate produce the same functionality or meaning as the reference? Rate or explain.”
- Ask a Jury of LLM as a Judge Pool, and take majority voting. If there is a tie, a manual interruption is needed as human-in-the-loop to break the system and proceed next.

An overview of LLM Jury

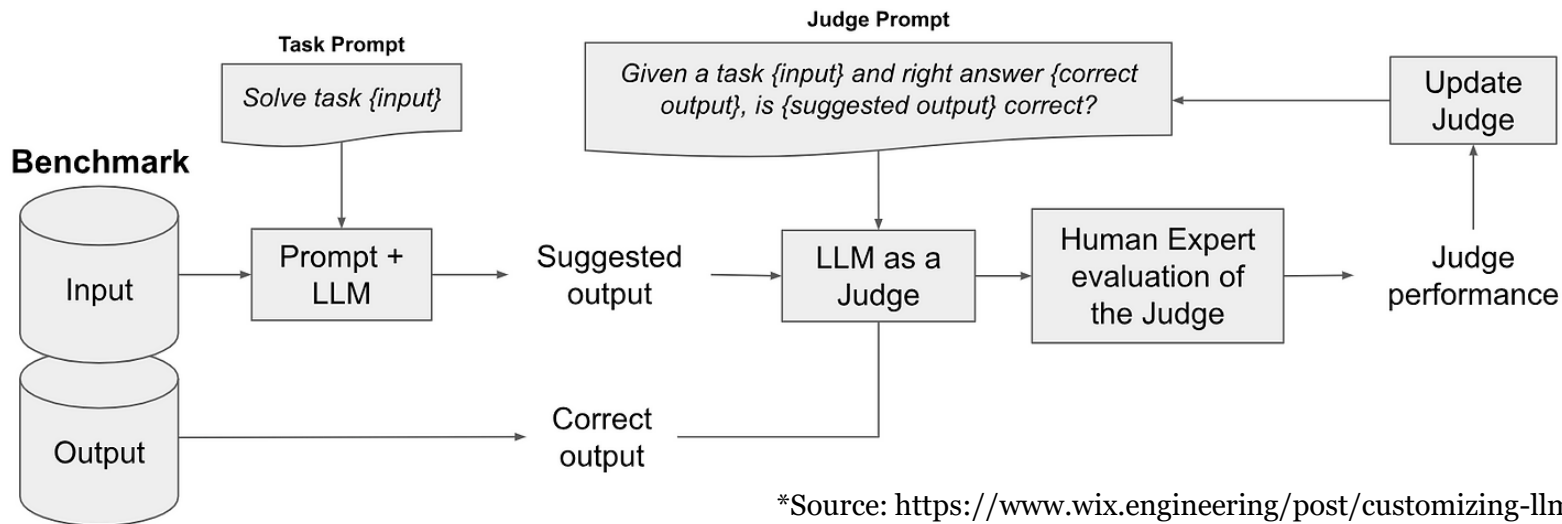


*Source: <https://arxiv.org/pdf/2410.09012>

Types of LLM judges

- **Pair wise comparison:** give LLM two responses and ask to choose the better one. This lets you compare models, prompts, or configurations to see which performs best.
- **Evaluation by criteria (reference-free):** ask the LLM to assess a response or conversation based on tone, clarity, correctness, or other dimensions.
- **Evaluation by criteria (reference-based):** provide extra context, like a source document or reference, and ask the LLM to score the response.

Development process for LLM-as-a-Judge



*Source: <https://www.wix.engineering/post/customizing-llms-for-enterprise-data-using-domain-adaptation-the-wix-journey>

Limitations of LLM-as-a-Judge: Biases

- **Position Bias:** An LLM may prefer a response simply because it appears earlier or later in a sequence.
- **Verbosity Bias:** It may unfairly equate longer responses with higher quality, regardless of content.
- **Authority Bias:** The LLM can be unduly influenced by references to authoritative sources rather than evaluating objectively.
- **Diversity Bias:** Judgments can be skewed by identity-related markers like gender or ethnicity.
- **Self-enhancement Bias:** An LLM judge may favor outputs from the same model or family of models, leading to a lack of impartiality.

Other Limitations of LLM-as-a-Judge

- **Difficulty with complex tasks:** Evaluating tasks that require complex reasoning, creative thinking, or practical expertise remains challenging for LLMs.
- **Lack of knowledge and context:** An LLM may lack the specific or proprietary context needed for accurate judgment.
- **Misleading confidence:** They can be "fooled" by answers that sound convincing but are incorrect, giving a false sense of confidence in the quality of an output.
- **Need for human oversight:** Because of these limitations, using LLM-as-a-judge requires a human to evaluate the judge itself and provide ground truth, especially for complex or critical tasks.

End of Simantic Similarity Score